

CSS

Professional Complete Guide

[Selectors](#) · [Box Model](#) · [Flexbox](#) · [Grid](#) · [Variables](#) · [Animations](#) · [Responsive](#) · [Advanced](#)

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Chapter 1: Selectors & Specificity

Every selector type and how the cascade works

Selector Reference

```

/* ■■■ Basic Selectors ■■■ */
* { box-sizing: border-box; } /* universal */
p { color: navy; } /* element */
.btn { padding: 8px 16px; } /* class */
#hero { height: 100vh; } /* id */

/* ■■■ Combinators ■■■ */
nav a { color: white; } /* descendant */
ul > li { list-style: disc; } /* direct child */
h2 + p { margin-top: 0; } /* adjacent sibling */
h2 ~ p { color: gray; } /* general sibling */

/* ■■■ Attribute Selectors ■■■ */
[href] { color: blue; } /* has attribute */
[type="email"] { border: blue; } /* equals */
[class~="btn"] { /* contains word */ }
[lang="en"] { /* equals or starts with en- */ }
[href^="https"] { /* starts with */ }
[href$=".pdf"] { /* ends with */ }
[href*="google"] { /* contains */ }

/* ■■■ Pseudo-classes ■■■ */
a:hover, a:focus { color: red; }
a:visited { color: purple; }
a:active { color: orange; }
input:focus { outline: 2px solid blue; }
input:focus-visible { /* keyboard focus only */ }
input:disabled { opacity: 0.5; }
input:checked { accent-color: blue; }
input:valid { border-color: green; }
input:invalid { border-color: red; }
input:placeholder-shown { border-style: dashed; }

li:first-child { font-weight: bold; }
li:last-child { border-bottom: none; }
li:nth-child(2n) { background: #f5f5f5; } /* even */
li:nth-child(3n+1) { /* 1st, 4th, 7th... */ }
li:nth-last-child(2) { /* 2nd from end */ }
p:first-of-type { font-size: 1.1em; }
p:only-child { text-align: center; }
:not(.special) { opacity: 0.7; }
:is(h1,h2,h3) { font-family: serif; } /* :is() */
:where(h1,h2,h3) { /* same but 0 specificity */ }
:has(img) { border: 1px solid; } /* parent selector */

/* ■■■ Pseudo-elements ■■■ */
p::first-letter { font-size: 3em; float: left; }
p::first-line { font-weight: bold; }
.card::before { content: "→ "; color: blue; }
.card::after { content: ""; display: block; clear: both; }
::selection { background: yellow; color: black; }
::placeholder { color: #aaa; font-style: italic; }
::marker { color: blue; font-size: 1.2em; }
::backdrop { background: rgba(0,0,0,0.5); }

```

```
/* ■■■ Combinators ■■■ */  
nav a { color: white; } /* descendant */  
ul > li { list-style: disc; } /* direct child */  
h2 + p { margin-top: 0; } /* adjacent sibling */  
h2 ~ p { color: gray; } /* general sibling */
```

```
/* ■■■ Attribute Selectors ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ */  
[href] { color: blue; } /* has attribute */  
[type="email"] { border: blue; } /* equals */  
[class~="btn"] { /* contains word */ }  
[lang|"en"] { /* equals or starts with en- */ }  
[href^="https"] { /* starts with */ }  
[href$=".pdf"] { /* ends with */ }  
[href*"google"] { /* contains */ }
```

[illegible]

```
li:first-child { font-weight: bold; }
li:last-child { border-bottom: none; }
li:nth-child(2n) { background: #f5f5f5; } /* even */
li:nth-child(3n+1) { /* 1st, 4th, 7th... */ }
li:nth-last-child(2) { /* 2nd from end */ }
p:first-of-type { font-size: 1.1em; }
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:not(.special) { opacity: 0.7; }
:is(h1,h2,h3) { font-family: serif; } /* :is() */
:where(h1,h2,h3) { /* same but 0 specificity */ }
:has(img) { border: 1px solid; } /* parent selector */
```

```
/* ■■■ Pseudo-elements ■■■■ * /  
p::first-letter { font-size: 3em; float: left; }  
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.card::before { content: "→ "; color: blue; }  
.card::after { content: ""; display: block; clear: both; }  
::selection { background: yellow; color: black; }  
::placeholder { color: #aaa; font-style: italic; }  
::marker { color: blue; font-size: 1.2em; }  
::backdrop { background: rgba(0,0,0,0.5); }
```

Specificity & Cascade

```
/* Specificity: (inline, id, class, element) */,
/* 0,0,0,1 */ p { }
/* 0,0,1,0 */ .btn { }
/* 0,0,1,1 */ p.intro { }
/* 0,1,0,0 */ #header { }
/* 0,1,1,1 */ #nav .link:hover { }
/* 1,0,0,0 */ style="color:red" (inline)
/* !important overrides everything – use sparingly */
.btn { color: red !important; }

/* Cascade layers (CSS @layer – modern approach) */
@layer reset, base, components, utilities;

@layer reset { *, *::before, *::after { margin:0; padding:0; } }
@layer base { body { font-family: system-ui; } }
@layer components { .btn { padding: 0.5rem 1rem; } }
@layer utilities { .mt-4 { margin-top: 1rem; } }
```

Chapter 2: Colors & Typography

Color systems, fonts, text properties

Color Systems

```
/* Named colors (140 available) */
color: tomato; background: lavender;

/* Hex */
color: #264DE4; /* 6-digit */
color: #26f; /* 3-digit shorthand = #2266ff */
color: #264DE480; /* 8-digit with alpha (80 = 50% opacity) */

/* RGB / RGBA */
color: rgb(38, 77, 228);
color: rgba(38, 77, 228, 0.5);
color: rgb(38 77 228 / 50%); /* modern syntax */

/* HSL / HSLA — most intuitive for designers */
color: hsl(228, 75%, 52%); /* hue saturation lightness */
color: hsla(228, 75%, 52%, 0.8);
color: hsl(228 75% 52% / 80%); /* modern */

/* HWB (hue whiteness blackness) */
color: hwb(228 10% 5%);

/* LCH / LAB (perceptually uniform) */
color: lch(50% 80 230);

/* CSS Custom Properties for theming */
:root {
  --color-primary: hsl(228 75% 52%);
  --color-secondary: hsl(168 75% 40%);
  --color-text: hsl(0 0% 13%);
  --color-bg: hsl(0 0% 98%);
}

[data-theme="dark"] {
  --color-text: hsl(0 0% 90%);
  --color-bg: hsl(0 0% 10%);
}

.btn { background: var(--color-primary); }
```

Typography

```

/* Font stack */
body {
font-family: "Inter", "Segoe UI", system-ui, -apple-system, sans-serif;
font-size: 16px; /* sets 1rem = 16px */
line-height: 1.6; /* unitless – best practice */
color: var(--color-text);
}

/* Font loading */
@import url("https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700&display=swap");

@font-face {
font-family: "MyFont";
src: url("myfont.woff2") format("woff2"),
url("myfont.woff") format("woff");
font-weight: 400;
font-style: normal;
font-display: swap; /* show fallback while loading */
}

/* Variable fonts */
@font-face {
font-family: "Roboto Flex";
src: url("RobotoFlex.woff2") format("woff2");
font-weight: 100 900; /* entire weight range */
}
. bold { font-variation-settings: "wght" 700; }
. condensed { font-variation-settings: "wdth" 75; }

/* Text properties */
h1 {
font-size: clamp(1.8rem, 4vw, 3rem); /* fluid typography */
font-weight: 700;
letter-spacing: -0.02em;
line-height: 1.2;
text-transform: uppercase;
font-feature-settings: "kern" 1, "liga" 1;
}
p {
text-align: justify;
text-indent: 1.5em;
hyphenate: auto;
hanging-punctuation: first;
}
. truncate {
overflow: hidden;
white-space: nowrap;
text-overflow: ellipsis;
}
. line-clamp {
display: -webkit-box;
-webkit-line-clamp: 3;
-webkit-box-orient: vertical;
overflow: hidden;
}

```

Chapter 3: Box Model & Sizing

margin, padding, border, overflow

Box Model

```
/* Box-sizing best practice */
*, *::before, *::after { box-sizing: border-box; }

.card {
  /* Content dimensions */
  width: 320px;
  min-width: 200px; max-width: 100%;
  height: auto;
  min-height: 200px;
  aspect-ratio: 16 / 9; /* maintain ratio */

  /* Padding – space inside border */
  padding: 24px; /* all */
  padding: 16px 24px; /* top/bottom, left/right */
  padding: 8px 16px 24px 12px; /* top right bottom left */
  padding-block: 16px; /* logical: top + bottom */
  padding-inline: 24px; /* logical: left + right */

  /* Border */
  border: 2px solid #264DE4;
  border-radius: 12px;
  border-radius: 50%; /* circle */
  border-radius: 8px 8px 0 0; /* top only */
  border-top: 4px solid gold;
  outline: 3px solid blue; /* does NOT affect layout */
  outline-offset: 4px;

  /* Margin – space outside border */
  margin: 0 auto; /* center horizontally */
  margin-block: 2rem; /* top + bottom */

  /* Shadow */
  box-shadow: 0 4px 16px rgba(0,0,0,0.12),
    0 1px 4px rgba(0,0,0,0.08);
  box-shadow: inset 0 2px 4px rgba(0,0,0,0.1); /* inner */

  /* Overflow */
  overflow: hidden; /* auto, scroll, visible, clip */
  overflow-x: auto; overflow-y: hidden;
  overflow-wrap: break-word; /* long words */
}
```

Chapter 4: Flexbox — Complete

Every flexbox property explained

Flex Container Properties

```
.container {
display: flex; /* or inline-flex */

/* Direction */
flex-direction: row; /* row | row-reverse | column | column-reverse */

/* Wrapping */
flex-wrap: nowrap; /* nowrap | wrap | wrap-reverse */

/* Shorthand */
flex-flow: row wrap; /* flex-direction flex-wrap */

/* Main axis alignment (horizontal in row) */
justify-content: flex-start;
/* flex-start | flex-end | center */
/* space-between | space-around | space-evenly */

/* Cross axis alignment (vertical in row) */
align-items: stretch;
/* stretch | flex-start | flex-end | center | baseline */

/* Multi-line cross axis */
align-content: normal;
/* normal | flex-start | flex-end | center */
/* space-between | space-around | space-evenly | stretch */

/* Gap between items */
gap: 16px; /* row-gap column-gap */
gap: 12px 24px; /* row-gap column-gap */
}
```

Flex Item Properties


```
.item {  
/* Flex grow – share available space */  
flex-grow: 0; /* default: dont grow */  
flex-grow: 1; /* grow to fill space */  
flex-grow: 2; /* grow twice as much as flex-grow:1 */  
  
/* Flex shrink – shrink when needed */  
flex-shrink: 1; /* default: can shrink */  
flex-shrink: 0; /* do not shrink */  
  
/* Flex basis – starting size */  
flex-basis: auto; /* use width/height */  
flex-basis: 0; /* ignore content size */  
flex-basis: 200px;  
  
/* Shorthand */  
flex: 0 1 auto; /* default */  
flex: 1; /* 1 1 0 – grow and shrink equally */  
flex: auto; /* 1 1 auto */  
flex: none; /* 0 0 auto – rigid */  
  
/* Self alignment – overrides align-items */  
align-self: auto | flex-start | flex-end | center | stretch;  
  
/* Order – visual reordering (does not affect DOM) */  
order: 0; /* default */  
order: -1; /* move to front */  
order: 99; /* move to back */  
}  
  
/* ■■■ Common Flexbox Patterns ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ */  
  
/* Centered content */  
.center { display:flex; justify-content:center; align-items:center; }  
  
/* Navigation bar */  
.navbar { display:flex; justify-content:space-between; align-items:center; }  
  
/* Card grid */  
.cards { display:flex; flex-wrap:wrap; gap:1rem; }  
.cards > * { flex: 1 1 280px; } /* responsive auto-columns */  
  
/* Push last item to end */  
.list-item:last-child { margin-left: auto; }  
  
/* Equal height columns */  
.cols { display:flex; align-items:stretch; }
```

Chapter 5: CSS Grid — Complete

Two-dimensional layout mastery

Grid Container

```
.grid {
display: grid; /* or inline-grid */

/* Columns & Rows */
grid-template-columns: 200px 1fr 1fr; /* fixed + flexible */
grid-template-columns: repeat(3, 1fr); /* 3 equal cols */
grid-template-columns: repeat(auto-fit, minmax(250px, 1fr)); /* responsive */
grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
grid-template-rows: 80px auto 60px;
grid-auto-rows: minmax(100px, auto); /* implicit rows */
grid-auto-columns: 200px;
grid-auto-flow: row; /* row | column | dense */

/* Gap */
gap: 16px; /* row-gap: 16px; column-gap: 16px */
gap: 12px 24px; /* row column */

/* Named template areas */
grid-template-areas:
"header header header"
"sidebar main main"
"sidebar footer footer";

/* Alignment of all items */
justify-items: stretch; /* start|end|center|stretch */
align-items: stretch;
place-items: center; /* align-items justify-items */

/* Alignment of the grid itself in container */
justify-content: start;
align-content: start;
place-content: center;
}
```

Grid Item Placement

```
.header { grid-area: header; } /* named area */
.sidebar { grid-area: sidebar; }
.main { grid-area: main; }
.footer { grid-area: footer; }

/* Explicit placement */
.item {
grid-column: 1 / 3; /* col 1 to col 3 */
grid-column: 1 / span 2; /* span 2 cols from col 1 */
grid-column: 2 / -1; /* col 2 to last line */
grid-row: 1 / 3;
grid-row: span 2;

/* Self alignment */
justify-self: center;
align-self: end;
place-self: center end;
}

/* ■■■ Responsive Grid Examples ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ */

/* Auto-responsive cards – no media queries needed! */
.card-grid {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
gap: 24px;
}

/* Classic page layout */
@media (min-width: 768px) {
.page {
display: grid;
grid-template-columns: 250px 1fr;
grid-template-rows: auto 1fr auto;
grid-template-areas:
    "header header"
    "sidebar main"
    "footer footer";
min-height: 100vh;
}
}
```

Chapter 6: Custom Properties & Animations

Variables, calc(), transitions, keyframes

CSS Custom Properties (Variables)

```
/* Define variables in :root for global scope */
:root {
  /* Colors */
  --clr-primary: hsl(228, 75%, 52%);
  --clr-secondary: hsl(168, 75%, 40%);
  --clr-bg: hsl(0, 0%, 98%);
  --clr-text: hsl(0, 0%, 13%);
  --clr-border: hsl(0, 0%, 87%);

  /* Spacing scale */
  --space-xs: 0.25rem; --space-sm: 0.5rem;
  --space-md: 1rem; --space-lg: 2rem;
  --space-xl: 4rem;

  /* Typography */
  --font-body: "Inter", system-ui, sans-serif;
  --font-mono: "Fira Code", "Courier New", monospace;
  --text-base: 1rem;
  --text-scale: 1.25; /* major third */

  /* Radius & Shadow */
  --radius-sm: 4px; --radius-md: 8px; --radius-lg: 16px;
  --shadow-sm: 0 1px 3px rgba(0,0,0,0.12);
  --shadow-md: 0 4px 16px rgba(0,0,0,0.12);

  /* Transitions */
  --transition: 200ms ease;
}

/* Dark mode */
@media (prefers-color-scheme: dark) {
  :root {
    --clr-bg: hsl(220, 13%, 12%);
    --clr-text: hsl(220, 13%, 90%);
  }
}

/* calc() — mix units */
.sidebar { width: calc(100% - 300px); }
.fluid-box { padding: calc(1rem + 2vw); }
.grid-col { width: calc(100% / var(--cols) - var(--gap)); }

/* clamp(min, preferred, max) */
h1 { font-size: clamp(1.5rem, 4vw, 3rem); }
.container { width: clamp(320px, 90%, 1200px); }
```

Transitions & Animations

```

/* Transitions */
.btn {
background: var(--clr-primary);
transition: background 300ms ease,
transform 200ms ease,
box-shadow 300ms ease;
}
.btn:hover {
background: hsl(228 75% 40%);
transform: translateY(-2px);
box-shadow: var(--shadow-md);
}

/* Timing functions */
/* ease | linear | ease-in | ease-out | ease-in-out */
/* cubic-bezier(0.4, 0, 0.2, 1) – Material Design easing */
/* steps(4, end) – stepped animation */

/* @keyframes */
@keyframes fadeInUp {
from { opacity: 0; transform: translateY(24px); }
to { opacity: 1; transform: translateY(0); }
}

@keyframes spin {
to { transform: rotate(360deg); }
}

@keyframes pulse {
0%, 100% { transform: scale(1); }
50% { transform: scale(1.05); }
}

/* Animation shorthand */
.hero-text {
animation: fadeInUp 600ms cubic-bezier(0.4,0,0.2,1) both;
animation-delay: 200ms;
}

.loader {
animation: spin 1s linear infinite;
}

/* animation properties */
/* animation-name, animation-duration, animation-timing-function */
/* animation-delay, animation-iteration-count, animation-direction */
/* animation-fill-mode: none|forwards|backwards|both */
/* animation-play-state: running|paused */

/* Respect user preferences */
@media (prefers-reduced-motion: reduce) {
*, *::before, *::after {
animation-duration: 0.01ms !important;
transition-duration: 0.01ms !important;
}
}

```

Chapter 7: Responsive Design

Media queries, container queries, fluid layouts

Media Queries

```
/* Mobile-first approach (recommended) */
/* Base styles: mobile */
body { font-size: 14px; }
.grid { grid-template-columns: 1fr; }

/* Tablet and up */
@media (min-width: 640px) {
body { font-size: 15px; }
.grid { grid-template-columns: repeat(2, 1fr); }
}

/* Desktop and up */
@media (min-width: 1024px) {
body { font-size: 16px; }
.grid { grid-template-columns: repeat(3, 1fr); }
.container { max-width: 1200px; margin: 0 auto; }
}

/* Large desktop */
@media (min-width: 1440px) {
.container { max-width: 1400px; }
}

/* Other media features */
@media (orientation: landscape) { /* ... */ }
@media (prefers-color-scheme: dark) { /* ... */ }
@media (prefers-reduced-motion: reduce) { /* ... */ }
@media (hover: hover) { .btn:hover { /* ... */ } }
@media print { .no-print { display: none; } }

/* Range syntax (modern) */
@media (600px <= width <= 1200px) { /* tablet only */ }

/* Logical operators */
@media (min-width: 768px) and (max-width: 1024px) { }
@media screen, print { } /* comma = OR */
@media not print { }

/* Container queries (CSS Containment) */
.card-wrapper {
container-type: inline-size;
container-name: card;
}

@container card (min-width: 400px) {
.card { flex-direction: row; }
}
```

Chapter 8: Advanced CSS

clip-path, filters, blend modes, pseudo-elements

Transforms

```
.el {  
  /* 2D transforms */  
  transform: translate(50px, 20px);  
  transform: translateX(50px) translateY(20px);  
  transform: rotate(45deg);  
  transform: scale(1.5);  
  transform: scale(1.5, 0.8); /* x, y */  
  transform: skew(15deg, 10deg);  
  
  /* Chaining – applied right to left */  
  transform: rotate(45deg) scale(1.2) translateX(100px);  
  
  /* 3D transforms */  
  transform: rotateX(45deg) rotateY(30deg);  
  transform: perspective(500px) rotateY(30deg);  
  transform-origin: center center; /* pivot point */  
  transform-origin: top left;  
  perspective: 1000px; /* on parent for 3D children */  
  backface-visibility: hidden;  
}
```

Filters, Clip-path & Blend Modes

```
/* CSS Filters */
img.grayscale { filter: grayscale(100%); }
img.blur { filter: blur(4px); }
img.brightness { filter: brightness(1.5); }
.shadow { filter: drop-shadow(4px 4px 8px rgba(0,0,0,0.3)); }
.vintage { filter: sepia(60%) contrast(1.1) brightness(0.9); }

/* Backdrop filter (frosted glass) */
.glass {
background: rgba(255,255,255,0.1);
backdrop-filter: blur(12px) saturate(180%);
}

/* Clip-path */
.circle { clip-path: circle(50%); }
.ellipse { clip-path: ellipse(60% 40% at 50% 50%); }
.triangle { clip-path: polygon(50% 0%, 0% 100%, 100% 100%); }
.hero { clip-path: polygon(0 0, 100% 0, 100% 85%, 0 100%); }

/* Mix blend modes */
.overlay { mix-blend-mode: multiply; }
.lighten { mix-blend-mode: screen; }
.text { mix-blend-mode: overlay; }

/* Background blend mode */
.bg { background-blend-mode: multiply; }

/* Aspect ratio */
.video-wrapper { aspect-ratio: 16 / 9; }
.avatar { aspect-ratio: 1; }

/* object-fit for images/videos in a container */
img { width: 100%; height: 300px; object-fit: cover; object-position: center top; }
```


Chapter 9: Exercises

20 CSS exercises covering all topics

Exercises

FILL

Ex 1. What is the specificity value of: #nav .item:hover?

■ **Answer:** 0,1,1,1 – (0 inline, 1 id, 1 class+pseudo-class, 1 element... actually: id=1, class+pseudo=2, elements=0 = 0,1,2,0)

WRITE

Ex 2. Write CSS for a sticky header that stays at the top when scrolling.

■ **Answer:** `header { position: sticky; top: 0; z-index: 100; background: white; box-shadow: 0 2px 8px rgba(0,0,0,0.1); }`

TRUEFALSE

Ex 3. gap works in both Flexbox and CSS Grid. (True/False)

■ **Answer:** TRUE – gap (and row-gap, column-gap) work in both flex and grid containers.

WRITE

Ex 4. Write a CSS custom property system with primary/secondary colors and use them on a button.

■ **Answer:** `:root { --clr-primary: hsl(228,75%,52%); --clr-primary-dark: hsl(228,75%,38%); }
.btn { background: var(--clr-primary); color: white; } .btn:hover { background: var(--clr-primary-dark); }`

FILL

Ex 5. What CSS property and value creates a frosted glass effect?

■ **Answer:** `backdrop-filter: blur(12px);` – with a semi-transparent background

WRITE

Ex 6. Write a 3-column card grid that collapses to 1 column on mobile using Grid.

■ **Answer:** `.grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(280px,1fr)); gap: 24px; }`

FILL

Ex 7. What does clamp(1rem, 4vw, 3rem) do?

■ **Answer:** Sets font-size to minimum 1rem, preferred 4vw (fluid), maximum 3rem – fluid typography without media queries.

WRITE

Ex 8. Write a CSS keyframe animation that makes an element bounce.

■ **Answer:** `@keyframes bounce { 0%,100%{transform:translateY(0)} 50%{transform:translateY(-20px)} } .bounce { animation: bounce 0.6s ease infinite; }`

EXPLAIN

Ex 9. Explain the difference between display:none, visibility:hidden, and opacity:0.

■ **Answer:** display:none: removes from layout, not focusable. visibility:hidden: invisible but takes space, not focusable. opacity:0: invisible but takes space, IS focusable/clickable. For accessibility, use display:none for truly hidden content.

WRITE

Ex 10. Write CSS for a perfect circle avatar with a border.

■ **Answer:** `.avatar { width:80px; height:80px; border-radius:50%; border:3px solid white; box-shadow:0 0 0 2px #264DE4; object-fit:cover; }`

FILL

Ex 11. What CSS property do you set to prevent an element from shrinking in Flexbox?

■ **Answer:** `flex-shrink: 0;`

WRITE

Ex 12. Write a CSS Grid layout with header, sidebar, main, and footer using named areas.

■ **Answer:** `.page { display:grid; grid-template-columns:250px 1fr; grid-template-rows:60px 1fr 50px; grid-template-areas:'header header' 'sidebar main' 'footer footer'; } .header{grid-area:header;} .sidebar{grid-area:sidebar;} .main{grid-area:main;} .footer{grid-area:footer;}`

EXPLAIN

Ex 13. What is the CSS cascade and how does it work?

■ **Answer:** The cascade resolves conflicts between rules using: (1) Origin & importance (!important), (2) Specificity (inline > #id > .class > element), (3) Order (last rule wins if equal specificity). CSS Layers (@layer) also factor in.

WRITE

Ex 14. Write a smooth CSS transition for a card hover effect (lift and shadow).

■ **Answer:** `.card { transition: transform 200ms ease, box-shadow 200ms ease; } .card:hover { transform: translateY(-4px); box-shadow: 0 12px 32px rgba(0,0,0,0.15); }`

FILL

Ex 15. What is the difference between em and rem units?

■ **Answer:** em is relative to the parent element's font-size (compounds). rem is relative to the root (:root/html) font-size – consistent and predictable. Prefer rem for font sizes.

WRITE

Ex 16. Write a CSS-only responsive navigation that collapses on mobile using a checkbox hack.

■ **Answer:** `input#toggle{display:none} .menu{display:none}
input#toggle:checked~.menu{display:flex}
@media(min-width:768px){.menu{display:flex!important} label{display:none}}`

TRUEFALSE

Ex 17. z-index works on elements with position:static. (True/False)

■ **Answer:** FALSE – z-index only works on positioned elements (relative, absolute, fixed, sticky). Elements with position:static ignore z-index.

WRITE

Ex 18. Write CSS for a loading spinner using border and animation.

■ **Answer:** `.spinner{width:40px;height:40px;border:4px solid
#eee;border-top-color:#264DE4;border-radius:50%;animation:spin 0.8s linear infinite;}
@keyframes spin{to{transform:rotate(360deg)}}`

EXPLAIN

Ex 19. What is BEM naming convention and why is it useful?

■ **Answer:** BEM = Block__Element--Modifier. Example: `.card__title--highlighted`. Benefits: predictable specificity (class-only), self-documenting structure, avoids naming conflicts, easier to maintain large codebases.

WRITE

Ex 20. Write a dark mode stylesheet using CSS custom properties and prefers-color-scheme.

■ **Answer:** `:root{--bg:#fff;--text:#111;--border:#ddd;}
@media(prefers-color-scheme:dark){:root{--bg:#121212;--text:#e0e0e0;--border:#333;}}
body{background:var(--bg);color:var(--text);}`